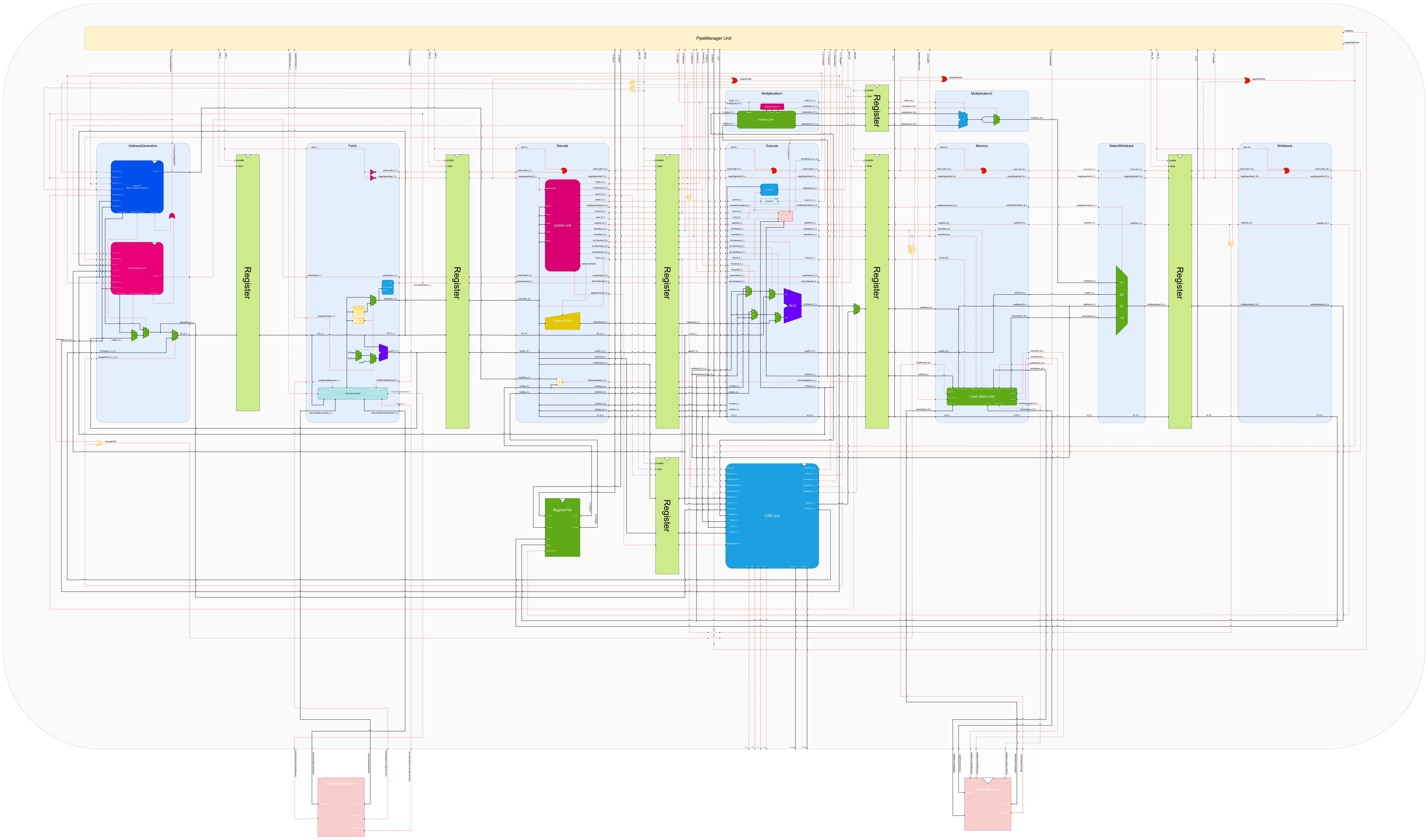
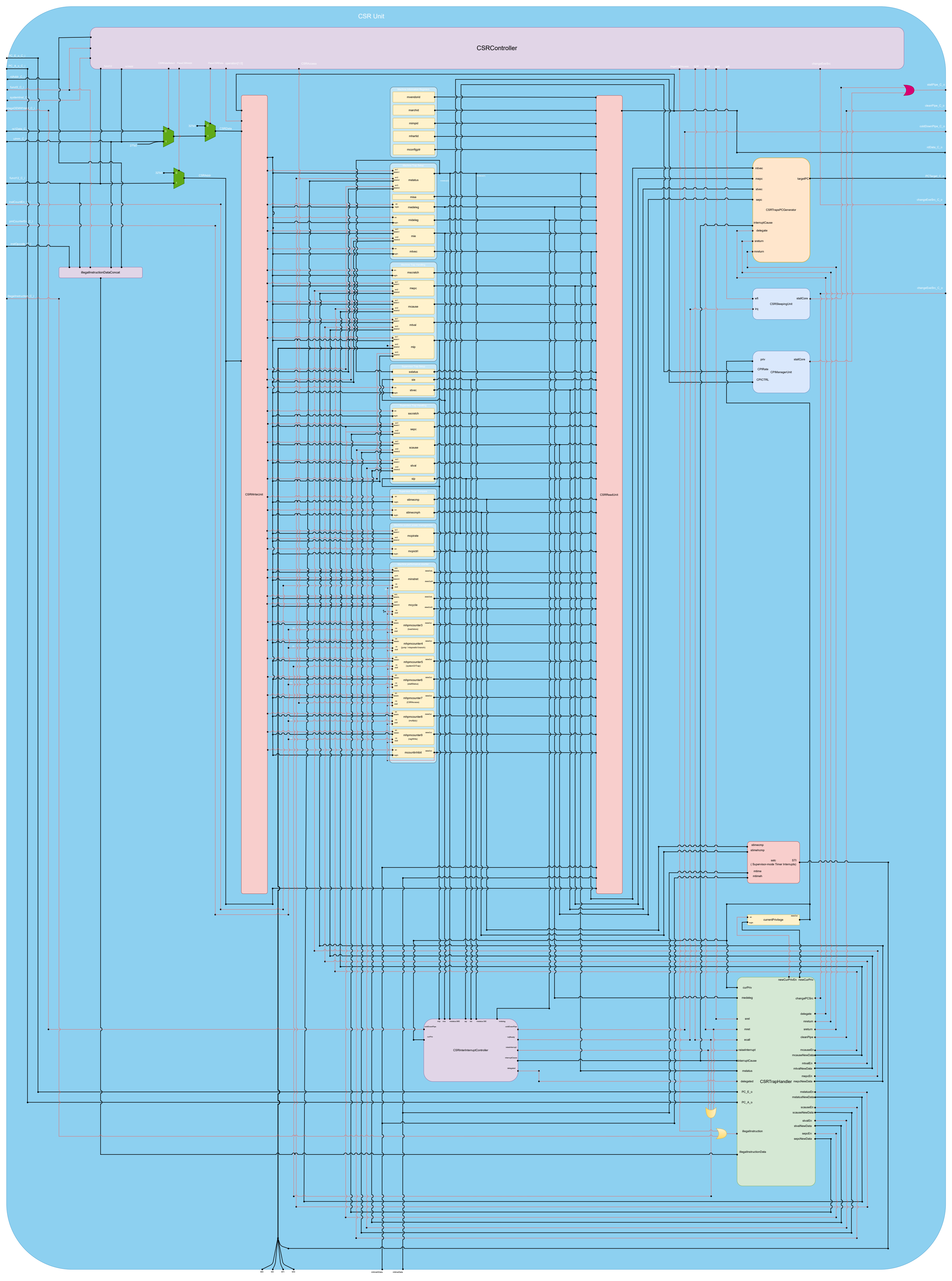
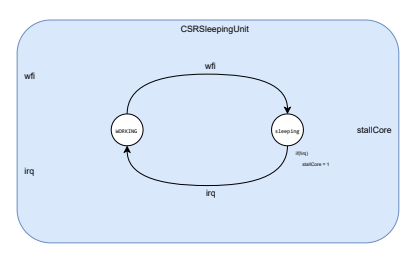
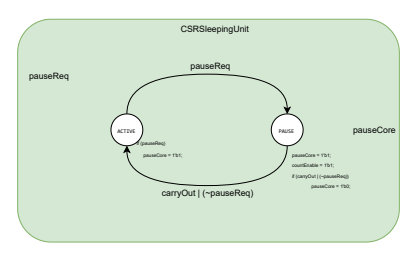

PipeManager Unit
AddressGeneration
Jump Unit
(Retrun Address prediction)
Branch Prediction Unit
Register
Fetch
InstructionLoadUnit
Decode
control unit
Extend Unit
RegisterFile
Execute
ALU
Multiplication1
wallace_tree
Multiplication2
Memory
Load Store Unit
SelectWriteback
Writeback
CSR Unit
stageDEValid
stageDEMValid
changePCSrc
enable
clear
Instruction Memory
Data Memory


CSR Unit
CSRController
IllegalInstructionDataConcat
CSRWriteUnit
CSRReadUnit
CSRTrapsPCGenerator
CSRSleepingUnit
CPManagerUnit
CSRInterruptController
CSRTrapHandler
currentPrivilege


CSRSleepingUnit
wfi
irq
stallCore


CSRSleepingUnit
pauseReq
pauseCore